BMJ Paediatrics Open

BMJ Paediatrics Open is committed to open peer review. As part of this commitment we make the peer review history of every article we publish publicly available.

When an article is published we post the peer reviewers' comments and the authors' responses online. We also post the versions of the paper that were used during peer review. These are the versions that the peer review comments apply to.

The versions of the paper that follow are the versions that were submitted during the peer review process. They are not the versions of record or the final published versions. They should not be cited or distributed as the published version of this manuscript.

BMJ Paediatrics Open is an open access journal and the full, final, typeset and author-corrected version of record of the manuscript is available on our site with no access controls, subscription charges or pay-per-view fees (http://bmjpaedsopen.bmj.com).

If you have any questions on BMJ Paediatrics Open's open peer review process please email info.bmjpo@bmj.com

BMJ Paediatrics Open

A Systematic Review of Complaints and/or Litigation Received by Neonatal Units

Journal:	*BMJ Paediatrics Open*
Manuscript ID	bmjpo-2021-001177
Article Type:	Original research
Date Submitted by the Author:	04-Jun-2021
Complete List of Authors:	Aiyengar, Apoorva; Homerton University Hospital NHS Foundation Trust, Neonatal Unit Aladangady, Narendra; Homerton University Hospital NHS Foundation Trust; Queen Mary University of London Morris, Tom; Homerton University Hospital NHS Foundation Trust, Neonatal Unit Bagshaw, Kaye; Homerton University Hospital NHS Foundation Trust, Newcombe Library
Keywords:	Neonatology, Health services research, Qualitative research

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

Confidential: For Review Only

Title: A Systematic Review of Complaints and/or Litigation Received by Neonatal Units

Authors: Apoorva Aiyengar[1], Tom Morris[1], Kaye Bagshaw[2], Narendra Aladangady[1,3]

1 Neonatal Unit, Homerton University Hospital NHS Foundation Trust, London, UK
2 Newcomb Library, Homerton University Hospital NHS Foundation Trust, London, UK
3 Child Health, Barts and the London School of Medicine and Dentistry, Queen Mary University of London, UK

Word count (excluding Abstract, Tables and References): 2435

Article type: Systematic Review (without meta analysis)

Corresponding author :
Professor Narendra Aladangady
Contact number 00442085105555
Email: n.aladangady@nhs.net
Postal Address
Neonatal Unit
Homerton University Hospital NHS Foundation Trust
Homerton Row, London, E9 6SR
UK

Confidential: For Review Only

Abstract- [248 words]

Importance*: Complaints and/or malpractice claims by family on the care of their babies in the neonatal unit is a pertinent issue. Beyond just the financial implications, it involves harm to babies and distress to parents.*

Objective: *The aim was to review published reports of complaints by family on the care of their babies in the neonatal units in order to understand the nature of these complaints and the areas of care that they relate to.*

Methods: *We considered articles in English which report on complaints made by families to neonatal units. We performed our structured search on AMED, CINAHL, EMBASE, EMCARE, SCOPUS and MEDLINE from January 2000 to December 2020. A total of 378 articles were appraised using eligibility criteria.*

Results: *A total of 12 articles were included. The most common category of complaint was delay in or incorrect diagnosis. Communication issues were highlighted as a significant category of complaints. The majority of the communication related claims were reported to be between the physicians and family. Factors implicated for clinical staff errors that resulted in parental complaint were lack of clinical and communication training, inadequate supervision of junior staff, work culture and hierarchy, not listening to family concerns and system failure.*

Conclusions: *Units should be encouraged to share information about such claims as they can facilitate shared learning. An understanding of human factor principles and its role in patient safety is also emphasized. This is to optimise patient outcomes and improve the neonatal unit experience for their families.*

Confidential: For Review Only

Confidential: For Review Only

Introduction [244 words]

In the United Kingdom, approximately 1 in 7 newborn babies (105,000) require specialist attention from neonatal units.[1] The advancement of neonatal medicine over the past 25 years has meant that smaller and more fragile babies are admitted to the Neonatal Intensive Care Unit (NICU) at earlier gestational ages.[2] Care is provided by specialist staff using highly developed equipment, and babies may undergo multiple medical or surgical procedures during a prolonged admission. Parents of babies on the neonatal unit describe an emotional and stressful journey during this period [3,4] and also afterwards, when some babies progress to experience long-term complications of their prematurity.[5] During this time, when the health and wellbeing of the baby hangs in the balance[6] and the trust that parents place in those taking care of their baby could not be more fragile, it stands to reason that medicolegal claims and complaints are commonly received by neonatal units.

A survey of American NICU physicians in practice for over 15 years showed that around 60% of physicians had faced at least one malpractice allegation.[7] However, there was no systematic review study published regarding complaints received at neonatal units. The aim of this systematic review was to evaluate such complaints. We have outlined the common themes across complaints received at different units and identified potential underlying causes for them. We have also addressed non-clinical aspects of care that make up complaints, which hold an equal value in improving the quality of patient care on NICU.

Methods/Literature Search [263 words]

This systematic review was conducted according to PRISMA guidelines.[8] The project protocol was registered with PROSPERO database.[9]

The literature search was performed by authors (KB and AA). Searches were performed on AMED, CINAHL, EMBASE, EMCARE, SCOPUS, MEDLINE and grey literature from January 2000 to December 2020. The strategy used to search the Medline database was as follows:

Confidential: For Review Only

("Neonatal unit" OR NICU OR "neonatal intensive care" OR SCBU OR "special care baby unit" OR Newborn OR Baby OR Babies OR Neonate OR Infant) AND (Complaint OR complaints OR litigation OR claim OR claims OR liability OR malpractice)

These terms were subsequently used in an equivalent way when searching the other databases. We restricted articles to a study population of infants and those reported in English. The initial list of articles obtained were screened using a title and abstract search by the first author (AA) and then full texts were reviewed by two independent reviewers (AA and TM) to ensure they met the eligibility criteria . Any disagreement or discrepancies were resolved with joint article review and discussion in meetings with the clinical members of research group (AA, TM and NA).In addition, a grey literature search was performed. We have excluded articles that reported complaints based on a single patient.

The articles included in our systematic review were collated and summarised in an excel spreadsheet. The number of neonatal cases, clinical category and category of complaints were extracted, and a descriptive thematic analysis was undertaken.

Patients or the public were not involved in the design, or conduct, or reporting, or dissemination plans of our research.

Confidential: For Review Only

Results [766 words]

Following our structured search, a total of 374 articles were obtained, with an additional 4 articles from a bibliography and grey literature search (Figure 1). Following the initial screening process of title and abstract review, 38 full text articles were examined for eligibility. A total of 12 studies were included in our systematic review and analysed. See Table 1 for a summary of the studies included.

The articles included in this systematic review reported on data collected in many different countries. Most of the articles reported data of neonatal complaints obtained from national databases (reflecting data from multiple centres), however two[10,11] studies reported data from single centres. Donn and Fanaroff [12] presented two separate legal cases involving the decision around providing therapeutic hypothermia as a treatment for hypoxic ischaemic injury in neonates.

Table 1: Summary of studies included in the systematic review of complaints received by neonatal units

Confidential: For Review Only

Authors [Ref No]	Study Methodology and country of origin	No of cases studied (Total)	No of complaint cases specifically involving neonatal team	Category of Clinical problem	Category of Complaint	Litigation and pay-outs
Mangurten et al, 2000[10][10]	• Retrospective review of cases (1972-1992) • Single centre – tertiary NICU • Review of medical and legal notes • Study conducted in USA	31	31	Seizure disorder (55%), hypoglycaemia (35%), pneumothorax (29%), HIE (23%), meconium aspiration (23%), asphyxia neonatorum (19%), IVH (16%), cerebral haemorrhage (16%), infantile apnoea (13%), hypotension (10%)	• Treatment error or delay (48%) • Missed/delay in diagnosis (16%) • Equipment misuse (6%) • General improper care (30%)	• 7 cases dismissed • 19 out of court settlement paid • 4 trial then damages paid • 1 unknown
Kaempf et al, 2016[17][11]	• Retrospective review of live births in single centre (April 1996- Dec 2013) of infants born at 22+0- 26+6 to identify resuscitation and outcome	606	3	• Management of extreme preterm infant against parental directive	• Lack of informed consent as parents felt they were pressured into making a decision without fully understanding the periviability counselling	• Unknown

Confidential: For Review Only

	• Highlight any cases with parental complaints or dissatisfaction expressed during counselling or formally via legal department • Study conducted in USA				• Decision for palliative comfort care chosen by the parents was reversed • Inconsistent communication with healthcare team	
Donn and Fanaroff, 2017 [20][12]	• Review article with 2 actual legal cases presented highlighting medicolegal issues around offering neuroprotective hypothermia • Legal cases in USA	2	2	Hypoxic ischaemic injury (HIE)	• Violation of the standard of care by not offering neuroprotective hypothermia (NPH) for suspected HIE • Offering NPH in an infant with an unknown cause of neurological injury leads to presupposition of an acute intrapartum event (which might not be the cause of the presentation)	• Unknown
Muniraman et al, 2017[11][13]	• Retrospective review of cases from Westlaw database (1980-2016) • National legal database • Review of medical and legal notes • Study conducted in USA	15	7	Peripartum and immediate postdelivery management of infants born between 22+0-25+6 gestation	• Conflicts on resuscitation contrary to parental directives. • Miscommunication between families and providers.	• All cases were legally handled • 4 case verdicts in favour of professionals • 2 unknown verdict • 1 case withdrawn

Confidential: For Review Only

Hawdon et al, 2016 [12][14]	• Retrospective review of cases from NHS Litigation Authority Claims Management database (2002-2011) • Data extracted from the national database, letters of complaints and responses • Study conducted in UK	28	28	Hypoglycaemia in infants >36/40 gestation	• Failure to appropriately monitor sugar levels despite risk factors • Early discharge without feeding support • Inadequate safety netting/discharge plan • Failure to take account of maternal concerns • Delayed testing • Missed diagnosis • Delayed referral to medical team once diagnosis made • Delayed treatment	• 25 cases damages paid • 4 cases defence only costs paid • 1 case no cost
Ashcroft B, 2008 [13][15]	• Retrospective review of babies that had been admitted to the Neonatal unit with severe birth asphyxia • Collected over a set time period (Feb 2001-March 2002) from 7 maternity units • Cognitive interviewing technique of staff involved, the data from the interviews and case notes were presented to an expert panel who applied the Bolam test	37	26	Birth asphyxia requiring neonatal intensive care input	• Delay in resuscitation	• Unknown

Confidential: For Review Only

	to identify acceptable standard of care • Study conducted in UK					
Nguyen J et al, 2017 [14][16]	• Retrospective review of cases from Westlaw database (Jan 1975- Aug 2016) • National legal database • Review of medical and legal notes • Study conducted in USA	167	54	• Procedure complication (24%) • Retinopathy of prematurity (19%) • Resuscitation (13%) • Infection (13%) • Other (31%)	• Communication related allegation (35%- 19/54) • Of above, miscommunication specifically between physician and family (74%) including lack of informed consent, lack of disclosure, lack of anticipatory guidance and management against parental wishes	• Unknown
Rennie et al 2018 [15][17]	• Retrospective review of cases from NHS resolution database (2001-2011) • Data extracted from the national database, letters of claim and response • Study conducted in UK	20	20	• Neonatal jaundice and kernicterus	• Delay in diagnosis (measuring serum bilirubin levels despite appearing visibly jaundiced and having risk factors for neonatal jaundice eg family history and presence of rhesus incompatibility with positive dat cord test n= 8 • Lag time between recognition and readmission (range between 26-102 hours) n= 15 • System failures (lack of NICU beds and inappropriate wait times	• Total costs of handling all claims = £150.5 million

Confidential: For Review Only

					in A+E resulting in delayed treatment with phototherapy) n=2	
Fallahi et al, 2016 [16][18]	• Retrospective review of malpractice cases in Tehran 2012-2014 • Obtained from Medical Commissions Office, Tehran	53	Unknown (due to cross over between obstetric and neonatal management)	• Cerebral palsy/birth asphyxia • Respiratory distress syndrome • Sepsis • Procedure related • Seizure	• Delay in diagnosis • Delay in treatment	• Unknown
Fanos et al, 2012 [18][19]	• Retrospective review of malpractice claims (2005-2010) from a nationwide database in Italy • Studied neonatal claims specifically in the labour room, nursery and NICU. • Study conducted in Italy	661	191	• Death due to all causes including infectious, respiratory, cardiac, gastrointestinal, anoxic, other. • Permanent injury due to neurological, joint related, vision/hearing, malformative, infectious or other	• Error in diagnosis • Improper performance (incorrect action by doctor, also includes claims that are due to failure to supervise or monitor patient) • No medical misadventure (claim has legal merit, but no clear error or mishap was made by neonatologist) • Medication error	• Unknown

Confidential: For Review Only

Berglund et al, 2008 [19][20]	• Retrospective review of malpractice claims (1990-2005 nationwide insurance database) • Study conducted in Sweden • Reviewed the immediate post delivery management in medical case notes (obstetric and neonatal) of severely asphyxiated infants following presumed malpractice in labour • Authors highlighted failure events in conjunction with neonatal resuscitation specifically based on their neonatal resus guidelines	472	177	• Neonatal resuscitation of infants born with evidence of asphyxia born after 33 weeks of gestation	• Unsatisfactory resuscitation (delay in starting artificial ventilation, intubation or chest compressions) • Unsatisfactory drug administration (lack of adrenaline or correction of metabolic acidosis) • Late initiation of resuscitation (appropriate skilled person not present at delivery or delayed in arriving) • Resuscitation not interrupted (despite >15 mins of asystole, or not spontaneously breathing for >30 mins)	• Unknown
Zhou et al, 2019 [21][21]	• Retrospective analysis of cerebral palsy malpractice claims collected from 1999-2017 • National database in China	400 (Data on errors available in 281 cases)	199	Cerebral Palsy	• Failure of timely and effective neonatal resuscitation • Improper handling of neonates (lack of observation, inadequate treatment or poor diagnosis) • Medical management error (lost or tampered medical records, poorly documented notes, physician without qualification)	• Difficult to calculate for cases specific to neonatal error. Mean value of compensation of service provider was $73, 506

Confidential: For Review Only

Category of Clinical Problems

The complaints received by neonatal units related to neonates with diagnoses spanning a wide range of clinical problems including respiratory (e.g. meconium aspiration, pneumonia, pneumothorax), cardiac, metabolic (e.g. hypoglycaemia), haematological (e.g. jaundice) and neurological (hypoxic ischaemic injury (HIE), intraventricular haemorrhage (IVH) and cerebral palsy). The details of each are listed in Table 1. In some cases, the infants often had more than one diagnosis as reported by Mangurten et al.[10]

Nguyen et al[16] reported complaints relating to a procedural complication which occurred in 13 (24%) cases, and this was also reported in 6 cases (5.4%) of malpractice by Fallahi et al.[18] Examples of procedural complications included issues with intravenous line placement, resulting in ischaemia to digit, and also acquired laceration during the delivery and resuscitation period.[18] Several studies[11,13,15,16,20] reported complaints relating to neonatal resuscitation

Category of Complaints

Confidential: For Review Only

Table 2 summarises the main categories of complaints received by the neonatal units. The most frequently reported categories of complaints are of incorrect/delayed diagnosis [10,12,14,17–19,21]and incorrect/delayed treatment. [10,12,14,17,18,21]As different studies reported complaints variably, and on occasion more than one category was cited to each complaint, it is difficult to quantify this and directly compare these categories across all the cases in the systematic review. Delay in initiation of resuscitation, including delay in administration of emergency drugs, was reported in 5 of the 12 studies.[11,13,15,20,21] Inappropriate resuscitation against parental directive was reported in 3 studies.[11,13,20] Communication issues were cited in 5 studies.[11,13–16] Medication errors featured in 3 studies[19–21] and general improper care of infants were reported in 4 studies.[10,14,19,21]

Table 2: Categories of complaints received by neonatal units

Category of Complaint	Studies [ref] that have reported such claims
Delay or Incorrect Diagnosis	Mangurten et al, 2000[10] Hawdon et al, 2016 [14] Rennie et al 2018 [17] Fallahi et al, 2016 [18] Fanos et al, 2012 [19] Donn and Fanaroff, 2017[12] Zhou et al, 2019 [21]
Delay in or Incorrect Treatment (not including resuscitation)	Mangurten et al, 2000[10] Hawdon et al, 2016 [14] Rennie et al 2018 [17] Fallahi et al, 2016 [18] Donn and Fanaroff, 2017 [12]

Confidential: For Review Only

	Zhou et al, 2019 [21]
Delay in Resuscitation / emergency drugs	Muniraman et al, 2017[13] Ashcroft B, 2008 [15] Kaempf et al, 2016 [11] Berglund et al, 2008 [20] Zhou et al, 2019 [21]
Inappropriate Initiation/continuation of Resuscitation	Muniraman et al, 2017[13] Kaempf et al, 2016 [11] Berglund et al, 2008 [20]
Communication Issue	Muniraman et al, 2017[13] Hawdon et al, 2016[14] Ashcroft B, 2008 [15] Nguyen J et al, 2017 [16] Kaempf et al, 2016 [11]
Medication error	Fanos et al, 2012 [19] Berglund et al, 2008 [20] Zhou et al, 2019 [21]
General Improper Care	Mangurten et al, 2000[10] Hawdon et al, 2016 [14] Fanos et al, 2012 [19] Zhou et al, 2019 [21]
Equipment issue	Mangurten et al, 2000[10] Hawdon et al, 2016 [14] Rennie et al 2018 [17]
Service Issue (includes data/medical records loss)	Rennie et al 2018 [17] Zhou et al, 2019 [21]
Procedural complications	Nguyen J et al, 2017 [16] Fallahi et al, 2016 [18]

Confidential: For Review Only

Factors implicated

Some of the studies included did investigate contributing factors to errors resulting in complaints (See Table 3). Ashcroft[15] undertook a cognitive interview of staff, exploring various factors implicated in serious incidents and outstanding claims on the Labour ward. This paper reported that failing to take action, making the wrong decision, or a delayed decision, were implicated as human errors in all the cases of malpractice they presented. Interestingly, they reported that the most dangerous time for errors to happen was during the night shift (51% n= 19 cases), where there was reduced availability of onsite expert medical support. Moreover, the author suggests that there is a host of systemic and cultural factors that propagate errors of decision-making with individual doctors and nurses in maternity. One such cause was poor staffing of labour wards and forcing inexperienced staff members to work without adequate supervision and support. In particular, this paper highlighted how junior paediatricians received little training in infant resuscitation; nevertheless in 46% of cases (n= 17) they were the first to be called, performing unsupervised resuscitation, including on severely asphyxiated infants. They identified that in 78% of the cases, senior staff did not provide assistance to inexperienced staff (including junior paediatricians), and that the junior staff often felt unable to ask for help for fear of being labelled as 'unable to cope'.[15]

Issues relating to suboptimal communication were highlighted as a significant category of complaints, as noted above and in Table 1 and 2. However, we could also consider poor communication as contributary to errors

Confidential: For Review Only

resulting in malpractice claims. Nguyen et al[16] reported that 74% of communication-related claims against neonatologists related to the communication between the physicians and family. This mostly occurred during prenatal counselling and resuscitation, when parents felt unable to give informed consent (57% of cases) having received no anticipatory guidance (21%) due to suboptimal communication by the neonatologist. Communication-related allegations against physicians and medical teams were also reported, and though there is little detail given about this in the article, miscommunication between team members could directly impact on patient safety and be a potential cause for errors. Kaemf et al [11]also included cases of parents reporting lack of informed consent through miscommunication during prenatal counselling discussions.

Table 3: Factors implicated for complaint against neonatal units

Factors Implicated in Cause of Errors	Study
Inadequate supervision of junior colleagues in resuscitation setting - delay in senior arriving	Ashcroft B, 2008[15]
Lack of training of junior doctors in resuscitation	Ashcroft B, 2008[15]
Culture of work and hierarchy resulting in a fear of asking for help	Ashcroft B, 2008[15]
Errors due to lack of adequate on site expert medical assistance (generally on shifts after 5pm or over the night the shift)	Ashcroft B, 2008[15]
Not adequately listening to maternal or family concerns	Hawdon et al, 2016[14]
Reduced access to proper equipment in a timely fashion e.g. Lab testing of glucose vs near patient testing devices	Hawdon et al, 2016[14]
System failures e.g. Lack of cots in Neonatal Unit or on the ward contributing to delays in initiating time critical treatment (phototherapy)	Rennie et al 2018[17]
Lack of training in communication	Nguyen J et al, 2017[16]
Staff shortage and high workload	Ashcroft B, 2008[15]
Human factors- eg tiredness	Ashcroft B, 2008[15]

Confidential: For Review Only

Discussion [937 words]

The most frequent category of complaint seen was *delayed/incorrect diagnosis*, followed closely by *delayed/incorrect treatment*. Though our systematic review focussed on complaints received by neonatal units, we are able to draw parallels with research in the field of paediatrics. Sen et al[22] analysed the causes and nature of successful litigation claims involving childhood fatalities over a 7-year period in the UK received by the NHS litigation authority. The most common cause for error (45% of cases) identified was delayed or failed diagnosis, followed by delayed or failed treatment in 15% of cases. Communication errors were highlighted in 4% of their cases; however, poor communication plays a role in other categories, including *inappropriate medical advice resulting in delayed presentation*, which was noted to be the cause of error in 5% of cases. [22]

Communication issues were highlighted as a significant category of complaints in our study. Neonatologists and paediatricians should be trained to provide accurate and understandable *safety-netting* advice to parents and caregivers, including when to seek help should their child unexpectedly deteriorate. This is not just relevant in

Confidential: For Review Only

accident and emergency departments when discharging a child, but also on the postnatal ward, as highlighted by Hawdon et al[14] eg. counselling parents about poor feeding is vital in picking up serious medical problems such as hypoglycaemia, sepsis and jaundice. Effective safety-netting not only helps parents and caregivers feel supported and reassured, strengthening the relationship between doctor and family, but can be lifesaving if a child deteriorates acutely.

Litigation was not a main focus of this review, and in the papers included in this systematic review which reported *on* litigation, the exact costs incurred were not always reported. In 2019/20, maternity claims (including potential neonatal cases) made up 9% of the number of clinical negligence claims received by the UK National Health Service Litigation Authority (NHS LA), but represented 50% of the £4.8 billion total claim value.[23] A medical error which results in harm to a neonate is a tragedy itself and, though money received as part of a claim can be put forward to caring for the additional needs and support of the child, it is difficult to put a price on the emotional distress incurred by the family. The distress experienced by the medical professional involved is also not to be underestimated: There is little data available on this subject, however facing a medical malpractice suit can be a stressful and isolating experience for the medical professional implicated.[24]

Furthermore, errors leading to patient harm in neonatal medicine are very rarely due to the mistakes of a single person. As quoted by James Reason,[25] it is often due to a ‘systems failure’, where a series of errors occur and

Confidential: For Review Only

align together, much like the *holes* in a *Swiss cheese* model of system failure, resulting in patient harm. An awareness of human factor principles and how these influence patient safety is also vital. In neonatal medicine, tight teamwork is key across multiple situations such as newborn resuscitation. What has been found to be of importance within a successful team is succinct and standardised communication, especially under time pressures and also use of clear protocols to minimise harmful organisational failures.[26] As mentioned by Fanaroff and Goldsmith[24], 'to borrow shamelessly' from mistakes made in other units, learning from these, and implementing safety measures in our units is one way to enhance patient safety. Another area to consider is staff training with an emphasis on evidence-based guidelines. Simulation training is a relatively new learning tool that has gained popularity in the paediatric training curriculum in the UK.[27] With the aid of equipment and mannikins, scenarios are staged allowing doctors and nurses to practice working together in simulated situations. This was incorporated into the neonatal teaching programme at one UK neonatal Unit,[28] addressing neonatal-specific training issues, e.g. endotracheal tube fixation, and received unanimously positive feedback from attending staff.

Strength and Limitations

Confidential: For Review Only

To our knowledge, this systematic review is the first of its kind to report on complaints received by neonatal units. From our broad-based searches of different databases, we have captured data from units across the world. However, by restricting searches to articles in English, we may have missed reports from neonatal units that have not published their data in English. Given the immense progression of neonatal medicine over the last few decades, we were keen to restrict our searches to articles published from the year 2000 as we believe the lessons to be learnt from before this may not necessarily be relevant to our current practice.

We excluded single case reports from this study, which inevitably has impacted the total number of articles we have reviewed. By only including papers reporting complaints *indiscriminately* from whole centres or groups of organisations, we have been able to consider our data with a lower risk of selection bias.

Two of the papers[13,16] included reported on complaint data collected through the Westlaw database which is a national legal database in America. It is possible that there may be some reporting duplication as the data was collected during overlapping years 1980-2016. Muniraman et al[13] focussed on complaints relating to peripartum or immediate post-delivery management of extreme preterm infants (study population included only 22-26 weeks gestation) primarily concerning neonatal resuscitation. Nguyen et al[16] studied all communication related allegations in infants born from 22-36 weeks of gestation. This included complaints relating to resuscitation but also reported on complaints relating to other clinical categories including procedural

Confidential: For Review Only

complications, infection and retinopathy of prematurity. We made the decision to include both papers as complaints relating to several different clinical areas which would be more reflective of complaints received by all neonatal units.

Conclusions [62 words]

Complaints and litigation about patient care are received by neonatal units. The most frequent categories of complaint reported in our systematic review were delay in/incorrect diagnosis and delay in/incorrect treatment. Through analysing and publishing their complaints data, neonatal units can contribute to a wider, shared understanding, helping guide future training and practice in a way that champions patient safety within the discipline.

[163 words]

What is already known about this topic?

- In 2019/20, maternity claims (including neonatal cases) made up 9% of the number of clinical negligence claims received by the NHS litigation authority, but represented 50% of the total claim value. [23]

Confidential: For Review Only

- Analysis of successful litigation claims in childhood fatalities over a 7 year period in England (in the paediatric population) showed that the top reasons included delay in diagnosis followed by delay in treatment.[22]
- In America, 60% of NICU physicians who had been in practice for over 15 years had experienced a case of malpractice litigation[7]

What does this study add?

- First systematic review to explore complaints in the neonatal population
- Communication related allegations are a significant cause of complaints. This includes communication during preterm counselling, resuscitation, during a NICU admission as well as in situations on the post-natal ward.
- Factors implicated for clinical staff errors that resulted in parental complaint were inadequate supervision of junior staff, work culture and hierarchy resulting in a fear of asking for help, not listening to family concerns and system failure.

Additional Figure 1 to add to article (legend)

Figure 1: Prisma diagram[8] depicting the stages of our systematic review

Confidential: For Review Only

Confidential: For Review Only

References

1 RCPCH. National Neonatal Audit Programme Annual Report 2020-on 2019 data. 2020. https://www.rcpch.ac.uk/sites/default/files/2020-11/nnap_report_2020_final_pdf2.pdf (accessed 10 Jan 2021).

2 Costeloe KL, Hennessy EM, Haider S, *et al.* Short term outcomes after extreme preterm birth in England: Comparison of two birth cohorts in 1995 and 2006 (the EPICure studies). *BMJ (Online)* 2012;**345**. doi:10.1136/bmj.e7976

3 Court B, Way W. Parents ' experiences of neonatal care: A report on the findings from a national survey. *Picker Institute Europe* 2011.

4 Shaw C, Connabeer K, Drew P, *et al.* Initiating end-of-life decisions with parents of infants receiving neonatal intensive care. *Patient Education and Counseling* 2020;**103**. doi:10.1016/j.pec.2020.02.013

5 Moore T, Hennessy EM, Myles J, *et al.* Neurological and developmental outcome in extremely preterm children born in England in 1995 and 2006: The EPICure studies. *BMJ (Online)* 2012;**345**. doi:10.1136/bmj.e7961

6 Aladangady N, Shaw C, Gallagher K, *et al.* Short-Term outcome of treatment limitation discussions for newborn infants, a multicentre prospective observational cohort study. *Archives of Disease in Childhood: Fetal and Neonatal Edition* 2017;**102**. doi:10.1136/archdischild-2016-310723

7 Meadow W, Bell A, Lantos J. Physicians' experience with allegations of medical malpractice in the neonatal intensive care unit. *Pediatrics* 1997;**99**. doi:10.1542/peds.99.5.e10

8 Liberati A, Altman DG, Tetzlaff J, *et al.* The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate healthcare interventions: explanation and elaboration. *BMJ (Clinical research ed)* 2009;**339**. doi:10.1136/bmj.b2700

9 Aiyengar A, Morris T, Aladangady N. PROSPERO International prospective register of systematic reviews. 2020. https://www.crd.york.ac.uk/prospero/display_record.php?ID=CRD42020167053

10 Mangurten HH, Angst DB, See C, *et al.* Professional liability in a neonatal intensive care unit: A review of 20 years' experience. *Journal of Perinatology* 2000;**20**. doi:10.1038/sj.jp.7200364

11 Kaempf JW, Tomlinson MW, Tuohey J. Extremely premature birth and the choice of neonatal intensive care versus palliative comfort care: An 18-year single-center experience. *Journal of Perinatology* 2016;**36**. doi:10.1038/jp.2015.171

12 Donn SM, Fanaroff JM. Medico-legal implications of hypothermic neuroprotection in the newborn. Journal of Neonatal-Perinatal Medicine. 2018;**11**. doi:10.3233/NPM-181792

13 Muniraman H, Cascione M, Ramanathan R, *et al.* Medicolegal cases involving periviable births from a major United States legal database. *Journal of Maternal-Fetal and Neonatal Medicine* 2018;**31**. doi:10.1080/14767058.2017.1335704

14 Hawdon JM, Beer J, Sharp D, *et al.* Neonatal hypoglycaemia: Learning from claims. *Archives of Disease in Childhood: Fetal and Neonatal Edition* 2017;**102**. doi:10.1136/archdischild-2016-310936

15 Ashcroft B. Labour ward incidents and potential claims - Lessons learned from research. *Clinical Risk* 2008;**14**. doi:10.1258/cr.2008.080081

16 Nguyen J, Muniraman H, Cascione M, *et al.* Communication-related allegations against physicians caring for premature infants. *Journal of Perinatology* 2017;**37**. doi:10.1038/jp.2017.113

17 Rennie JM, Beer J, Upton M. Learning from claims: Hyperbilirubinaemia and kernicterus. *Archives of Disease in Childhood: Fetal and Neonatal Edition* Published Online First: 2018. doi:10.1136/archdischild-2017-314622

18 Fallahi M, Vaghefi S, Makhmalbaf M, *et al.* Analysis of Negligence in field of Neonatology in Medical Commissions Office in Tehran, Iran. *International Journal of Medical Toxicology and Forensic Medicine* 2016;**6**. doi:10.22037/ijmtfm.v6i3(Summer).10773

19 Fanos V, Tagliabue P, Greco L, *et al.* Neonatal malpractice claims in Italy: How big is the problem and which are the causes? *Journal of Maternal-Fetal and Neonatal Medicine* 2012;**25**. doi:10.3109/14767058.2011.622004

20 Berglund S, Norman M, Grunewald C, *et al.* Neonatal resuscitation after severe asphyxia - A critical evaluation of 177 Swedish cases. *Acta Paediatrica, International Journal of Paediatrics* 2008;**97**. doi:10.1111/j.1651-2227.2008.00803.x

21 Zhou L, Li H, Li C, *et al.* Risk management and provider liabilities in infantile cerebral palsy based on malpractice litigation cases. *Journal of Forensic and Legal Medicine* 2019;**61**. doi:10.1016/j.jflm.2018.11.010

22 Sen G, Keene J, Raine J. An analysis of successful litigation claims in childhood fatalities in England. *European Journal of Pediatrics* 2012;**171**. doi:10.1007/s00431-012-1793-6

23 National Health Service. Litigation Authority. *NHS Resolution Annual report and accounts 2019/20*.

24 Fanaroff JM, Goldsmith JP. The most common patient safety issues resulting in legal action against neonatologists. Seminars in Perinatology. 2019. doi:10.1053/j.semperi.2019.08.010

25 Reason J. Human error: Models and management. British Medical Journal. 2000;**320**. doi:10.1136/bmj.320.7237.768

26 Yamada NK, Catchpole K, Salas E. The role of human factors in neonatal patient safety. Seminars in Perinatology. 2019. doi:10.1053/j.semperi.2019.08.003

27 Grant DJ, Marriage SC. Training using medical simulation. Archives of Disease in Childhood. 2012;**97**. doi:10.1136/archdischild-2011-300592

For Peer Review Only

28 Sharma A. From evidence to implementation: Introducing neonatal simulation to a tertiary neonatal centre in the UK. *Open Journal of Pediatrics* 2013;**03**. doi:10.4236/ojped.2013.31002

Confidential: For Review Only

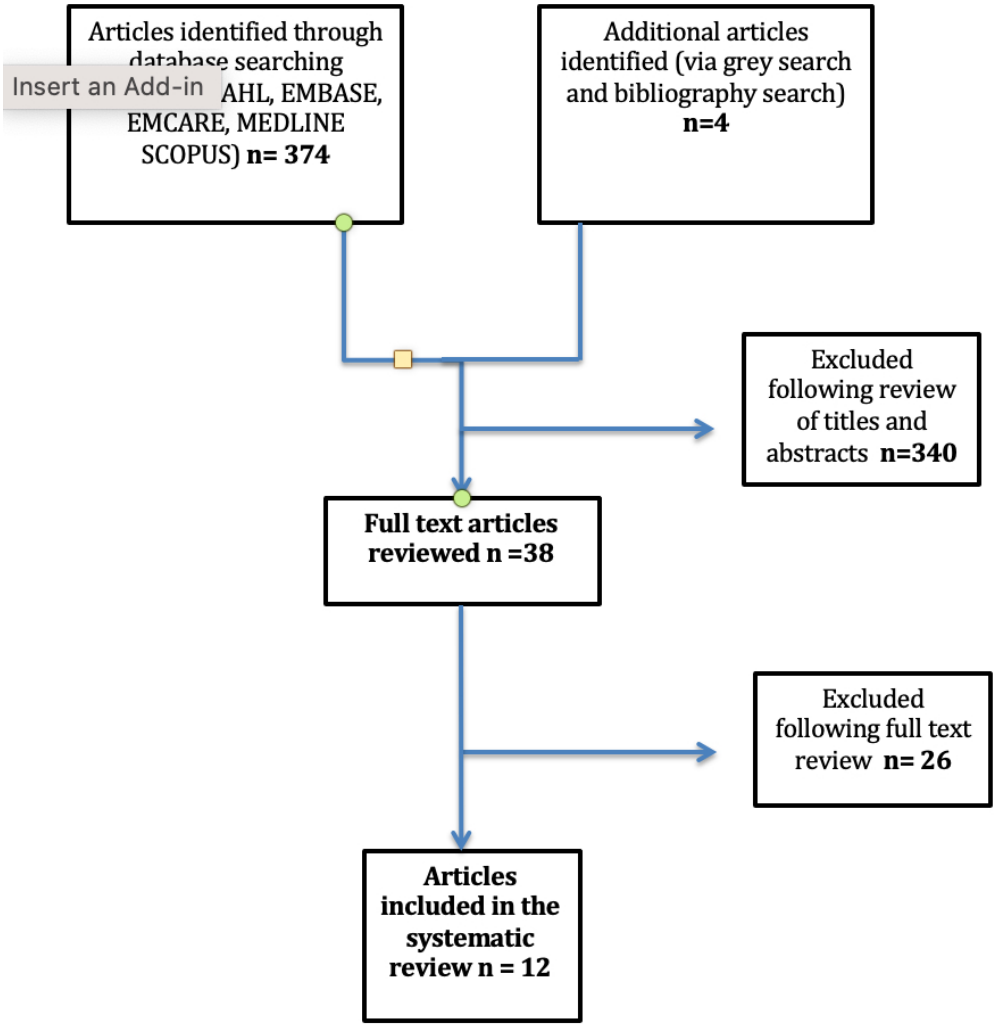


Figure 1: Prisma Diagram depicting the stages of our systematic review.

155x177mm (144 x 144 DPI)

BMJ Paediatrics Open

A Systematic Review of Medical Literature for Medico-legal Claims and Complaints Involving Neonates

Journal:	*BMJ Paediatrics Open*
Manuscript ID	bmjpo-2021-001177.R1
Article Type:	Original research
Date Submitted by the Author:	12-Jul-2021
Complete List of Authors:	Aiyengar, Apoorva; Homerton University Hospital NHS Foundation Trust, Neonatal Unit Morris, Tom; Homerton University Hospital NHS Foundation Trust, Neonatal Unit Bagshaw, Kaye; Homerton University Hospital NHS Foundation Trust, Newcombe Library Aladangady, Narendra; Homerton University Hospital NHS Foundation Trust; Queen Mary University of London
Keywords:	Neonatology, Health services research, Qualitative research

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

Confidential: For Review Only

Title: A Systematic Review of Medical Literature for Medico-legal Claims and Complaints Involving Neonates

Authors: Apoorva Aiyengar[1], Tom Morris[1], Kaye Bagshaw[2], Narendra Aladangady[1,3]

1 Neonatal Unit, Homerton University Hospital NHS Foundation Trust, London, UK
2 Newcomb Library, Homerton University Hospital NHS Foundation Trust, London, UK
3 Child Health, Barts and the London School of Medicine and Dentistry, Queen Mary University of London, UK

Word count (excluding Abstract, Tables and References): 2482

Article type: Systematic Review (without meta analysis)

Corresponding author :
Professor Narendra Aladangady
Contact number 00442085105555
Email: n.aladangady@nhs.net
Postal Address
Neonatal Unit
Homerton University Hospital NHS Foundation Trust
Homerton Row, London, E9 6SR
UK

Contributorship statement- AA was involved in planning of this systematic review, extracted the data from the studies and performed thematic analysis (together with NA and TM who have previous experience doing thematic analysis) AA prepared the manuscript and edited it. TM was involved in the planning of the systematic review and edited the manuscript. NA was the supervisor of this project and produced the idea for this systematic review. NA was involved in the planning of the project, analysis of data (thematic analysis) and editing the manuscript. KB performed the structured searches on the various databases in order to obtain data for the systematic review.

Confidential: For Review Only

Abstract- [253 words]

Importance*: Complaints and malpractice claims by familes on the care of their babies is a pertinent issue. Beyond just the financial implications, it involves harm to babies and distress to parents.*

Objective: *The aim was to review published reports of complaints by families on the care of their babies in the neonatal units in order to understand the nature of these complaints and the areas of care that they relate to.*

Methods: *We considered articles in English which report on complaints made by families to organisations providing neonatal care. We performed our structured search on AMED, CINAHL, EMBASE, EMCARE, SCOPUS and MEDLINE from January 2000 to December 2020. A total of 378 articles were appraised using eligibility criteria.*

Results: *A total of 12 articles were included. The most common category of complaint was delayed/incorrect diagnosis. Communication issues were highlighted as a significant category of complaints. The majority of such claims were between the physicians and families. Factors implicated for clinician's errors that resulted in complaints were lack of clinical and communication training, inadequate supervision of junior clinicians, work culture and hierarchy, not listening to families' concerns and system failure.*

Conclusions: *The most frequent categories of complaint reported in our systematic review were delayed/incorrect diagnosis and delayed/incorrect treatment. Organisations should be encouraged to share complaints data as it can facilitate shared learning. An understanding of human factor principles and its role in*

Confidential: For Review Only

patient safety is also emphasized in this report in order to optimise patient outcomes and improve experience for families requiring neonatal care.

Confidential: For Review Only

Introduction [247 words]

In the United Kingdom, approximately 1 in 7 newborn babies (105,000) require specialist attention from neonatal units.[1] The advancement of neonatal medicine over the past 25 years has meant that smaller and more fragile babies are admitted to the Neonatal Intensive Care Unit (NICU) at earlier gestational ages.[2] Care is provided by specialist staff using highly developed equipment, and babies may undergo multiple medical or surgical procedures during a prolonged admission. Parents of babies on the neonatal unit describe an emotional and stressful journey during this period [3,4] and also afterwards, when some babies progress to experience long-term complications of their prematurity.[5] During this time, when the health and wellbeing of the baby hangs in

the balance[6] and the trust that parents place in those taking care of their baby could not be more fragile, it stands to reason that medicolegal claims and complaints are commonly received by organisations that provide neonatal care.

A survey of American NICU physicians in practice for over 15 years showed that around 60% of physicians had faced at least one malpractice allegation.[7]However, there was no systematic review study published regarding complaints received at neonatal units. The aim of this systematic review was to evaluate such complaints. We have outlined the common themes across complaints received at different units and identified potential underlying causes for them. We have also addressed non-clinical aspects of care that make up complaints, which hold an equal value in improving the quality of patient care on NICU.

Methods/Literature Search [290 words]

Confidential: For Review Only

This systematic review was conducted according to PRISMA guidelines.[8] The project protocol was registered with PROSPERO database.[9]

The literature search was performed by authors (KB and AA). Searches were performed on AMED, CINAHL, EMBASE, EMCARE, SCOPUS, MEDLINE and grey literature from January 2000 to December 2020. The strategy used to search the Medline database was as follows:

("Neonatal unit" OR NICU OR "neonatal intensive care" OR SCBU OR "special care baby unit" OR Newborn OR Baby OR Babies OR Neonate OR Infant) AND (Complaint OR complaints OR litigation OR claim OR claims OR liability OR malpractice)

These terms were subsequently used in an equivalent way when searching the other databases. We restricted articles to a study population of infants and those reported in English. The initial list of articles obtained were screened using a title and abstract search by the first author (AA) and then full texts were reviewed by two independent reviewers (AA and TM) to ensure they met the eligibility criteria . Any disagreement or discrepancies were resolved with joint article review and discussion in meetings with the clinical members of research group (AA, TM and NA). In addition, a grey literature search was performed using Google and Google scholar search engines. We have excluded articles that reported complaints based on a single patient.

Confidential: For Review Only

The articles included in our systematic review were collated and summarised in an excel spreadsheet. The number of neonatal cases, clinical category and category of complaints were extracted. A descriptive thematic analysis was undertaken initially by AA and then finalised in discussion with clinical researchers with previous experience in thematic analysis (TM and NA.

Patients or the public were not involved in the design, or conduct, or reporting, or dissemination plans of our research.

Results [779 words]

Following our structured search, a total of 374 articles were obtained, with an additional 4 articles from a bibliography and grey literature search (Figure 1). Following the initial screening process of title and abstract review, 38 full text articles were examined for eligibility. A total of 12 studies were included in our systematic review and analysed. See Table 1a for a summary of the studies included. Please review Table 1b in the supplementary file for full details.

Confidential: For Review Only

Table 1a: Studies included in the systematic review

Authors [Ref No]	Country Of Origin of the Study	Study Methodology
Berglund et al, 2008 [10]	• Sweden	• Retrospective review of malpractice claims (1990-2005 nationwide insurance database) • Reviewed the immediate post delivery management in medical case notes (obstetric and neonatal) of severely asphyxiated infants following presumed malpractice in labour • Authors highlighted failure events in conjunction with neonatal resuscitation specifically based on their neonatal resus guidelines
Fallahi et al, 2016 [11]	• Tehran	• Retrospective review of malpractice cases in Tehran 2012-2014 • Data from the Medical Commissions office (National database)
Fanos et al, 2012 [12]	• Italy	• Retrospective review of malpractice claims (2005-2010) from a nationwide database in Italy • Studied neonatal claims specifically in the labour room, nursery and NICU.
Hawdon et al, 2016 [13]	• United Kingdom (UK)	• Retrospective review of cases from NHS Litigation Authority Claims Management database (2002-2011) • Data extracted from the national database, letters of complaints and responses
Muniraman et al, 2017[14]	• United States of	• Retrospective review of cases from Westlaw database (1980-2016) • National legal database

Confidential: For Review Only

	America (USA)	• Review of medical and legal notes
Nguyen J et al, 2017 [15]	• USA	• Retrospective review of cases from Westlaw database (Jan 1975- Aug 2016) • National legal database • Review of medical and legal notes
Rennie et al 2018 [16]	• UK	• Retrospective review of cases from NHS resolution database (2001-2011) • Data extracted from the national database, letters of claim and response
Zhou et al, 2019 [17]	• China	• Retrospective analysis of cerebral palsy malpractice claims collected from 1999-2017 • National database
Ashcroft B, 2008 [18]	• UK	• Retrospective review of babies that had been admitted to the Neonatal unit with severe birth asphyxia • Collected over a set time period (Feb 2001-March 2002) from 7 maternity units (Regional data) • Cognitive interviewing technique of staff involved, the data from the interviews and case notes were presented to an expert panel who applied the Bolam test to identify acceptable standard of care
Kaempf et al, 2016 [19]	• USA	• Retrospective review of live births in single centre (April 1996- Dec 2013) of infants born at 22+0- 26+6 to identify resuscitation and outcome • Highlight any cases with parental complaints or dissatisfaction expressed during counselling or formally via legal department
Mangurten et al, 2000[20]	• USA	• Retrospective review of cases (1972-1992) • Single centre – tertiary NICU • Review of medical and legal notes

Confidential: For Review Only

Donn and Fanaroff, 2017 [21]	• USA	• Review article with 2 actual legal cases presented highlighting medicolegal issues around offering neuroprotective hypothermia

The articles included in this systematic review reported on data collected in many different countries. Most of the articles reported data of neonatal complaints obtained from national databases (reflecting data from multiple centres), however two[19,20] studies reported data from single centres. Donn and Fanaroff [21] presented two separate legal cases involving the decision around providing therapeutic hypothermia as a treatment for hypoxic ischaemic injury in neonates.

Category of Clinical Problems (See table 1b-supplementary file)

The complaints received by neonatal units related to neonates with diagnoses spanning a wide range of clinical problems including respiratory (e.g. meconium aspiration, pneumonia, pneumothorax), cardiac, metabolic (e.g. hypoglycaemia), haematological (e.g. jaundice) and neurological (hypoxic ischaemic injury (HIE), intraventricular haemorrhage (IVH) and cerebral palsy). The details of each are listed in Table 1. In some cases, the infants often had more than one diagnosis as reported by Mangurten et al.[20]

Confidential: For Review Only

Nguyen et al[15] reported complaints relating to a procedural complication which occurred in 13 (24%) cases, and this was also reported in 6 cases (5.4%) of malpractice by Fallahi et al.[11] Examples of procedural complications included issues with intravenous line placement, resulting in ischaemia to digit, and also acquired laceration during the delivery and resuscitation period.[11] Several studies[10,14,15,18,19] reported complaints relating to neonatal resuscitation

Category of Complaints

Table 2 summarises the main categories of complaints received by the neonatal units. The most frequently reported categories of complaints are of incorrect/delayed diagnosis [11–13,16,17,20,21]and incorrect/delayed treatment. [11,13,16,17,20,21]As different studies reported complaints variably, and on occasion more than one category was cited to each complaint, it is difficult to quantify this and directly compare these categories across all the cases in the systematic review. Delay in initiation of resuscitation, including delay in administration of emergency drugs, was reported in 5 of the 12 studies.[10,14,17–19] Inappropriate resuscitation against parental directive was reported in 3 studies.[10,14,19] Communication issues were cited in 5 studies.[13–15,18,19] Medication errors featured in 3 studies[10,12,17] and general improper care of infants were reported in 4 studies.[12,13,17,20]

Table 2: Categories of complaints received by neonatal units

Confidential: For Review Only

Category of Complaint	Studies that have reported such claims
Delay or Incorrect Diagnosis	Mangurten et al, 2000[20] Hawdon et al, 2016 [13] Rennie et al 2018 [16] Fallahi et al, 2016 [11] Fanos et al, 2012 [12] Donn and Fanaroff, 2017[21] Zhou et al, 2019 [17]
Delay in or Incorrect Treatment (not including resuscitation)	Mangurten et al, 2000[20] Hawdon et al, 2016 [13] Rennie et al 2018 [16] Fallahi et al, 2016 [11] Donn and Fanaroff, 2017 [21] Zhou et al, 2019 [17]
Delay in Resuscitation / emergency drugs	Muniraman et al, 2017[14] Ashcroft B, 2008 [18] Kaempf et al, 2016 [19] Berglund et al, 2008 [10] Zhou et al, 2019 [17]
Inappropriate Initiation/continuation of Resuscitation	Muniraman et al, 2017[14] Kaempf et al, 2016 [19]

Confidential: For Review Only

	Berglund et al, 2008 [10]
Communication Issue	Muniraman et al, 2017[14] Hawdon et al, 2016[13] Ashcroft B, 2008 [18] Nguyen J et al, 2017 [15] Kaempf et al, 2016 [19]
Medication error	Fanos et al, 2012 [12] Berglund et al, 2008 [10] Zhou et al, 2019 [17]
General Improper Care	Mangurten et al, 2000[20] Hawdon et al, 2016 [13] Fanos et al, 2012 [12] Zhou et al, 2019 [17]
Equipment issue	Mangurten et al, 2000[20] Hawdon et al, 2016 [13] Rennie et al 2018 [16]
Service Issue (includes data/medical records loss)	Rennie et al 2018 [16] Zhou et al, 2019 [17]
Procedural complications	Nguyen J et al, 2017 [15] Fallahi et al, 2016 [11]

Confidential: For Review Only

Factors implicated

Some of the studies included did investigate contributing factors to errors resulting in complaints (See Table 3). Ashcroft[18] undertook a cognitive interview of clinicians, exploring various factors implicated in serious incidents and outstanding claims on the Labour ward. This paper reported that failing to take action, making the wrong decision, or a delayed decision, were implicated as human errors in all the cases of malpractice they presented. Interestingly, they reported that the most dangerous time for errors to happen was during the night shift (51% n= 19 cases), where there was reduced availability of onsite expert medical support. Moreover, the author suggests that there is a host of systemic and cultural factors that propagate errors of decision-making with individual doctors and nurses in maternity. One such cause was poor staffing of labour wards and forcing inexperienced clinicians to work without adequate supervision and support. In particular, this paper highlighted how junior paediatricians received little training in infant resuscitation; nevertheless in 46% of cases (n= 17) they were the first to be called, performing unsupervised resuscitation, including on severely asphyxiated infants. They identified that in 78% of the cases, senior clinicians did not provide assistance to inexperienced clinicians (including junior paediatricians), and that the juniors often felt unable to ask for help for fear of being labelled as 'unable to cope'.[18]

Confidential: For Review Only

Issues relating to suboptimal communication were highlighted as a significant category of complaints, as noted above and in Table 1 and 2. However, we could also consider poor communication as contributary to errors resulting in malpractice claims. Nguyen et al[15] reported that 74% of communication-related claims against neonatologists related to the communication between the doctors and family. This mostly occurred during prenatal counselling and resuscitation, when parents felt unable to give informed consent (57% of cases) having received no anticipatory guidance (21%) due to suboptimal communication by the neonatologist. Communication-related allegations between doctors and within medical teams were also reported, and though there is little detail given about this in the article, miscommunication between team members could directly impact on patient safety and be a potential cause for errors. Kaemf et al [19]also included cases of parents reporting lack of informed consent through miscommunication during prenatal counselling discussions.

Table 3: Factors implicated for complaint against neonatal units

Factors Implicated in Cause of Errors	**Study**
Inadequate supervision of junior colleagues in resuscitation setting - delay in senior arriving	Ashcroft B, 2008[18]
Lack of training of junior doctors in resuscitation	Ashcroft B, 2008[18]
Culture of work and hierarchy resulting in a fear of asking for help	Ashcroft B, 2008[18]
Errors due to lack of adequate on site expert medical assistance (generally on shifts after 5pm or over the night the shift)	Ashcroft B, 2008[18]
Not adequately listening to maternal or family concerns	Hawdon et al, 2016[13]
Reduced access to proper equipment in a timely fashion e.g. Lab testing of glucose vs near patient testing devices	Hawdon et al, 2016[13]
System failures e.g. Lack of cots in Neonatal Unit or on the ward contributing to delays in initiating time critical treatment (phototherapy)	Rennie et al 2018[16]
Lack of training in communication	Nguyen J et al, 2017[15]
Staff shortage and high workload	Ashcroft B, 2008[18]

Confidential: For Review Only

Human factors- eg tiredness	Ashcroft B, 2008[18]

Confidential: For Review Only

Discussion [1082 words]

The most frequent category of complaint seen was *delayed/incorrect diagnosis*, followed closely by *delayed/incorrect treatment*. Though our systematic review focussed on complaints and litigation received by organisations providing neonatal care, we are able to draw parallels with research in the field of paediatrics. Sen et al[22] analysed the causes and nature of successful litigation claims involving childhood fatalities over a 7-year period in the UK received by the NHS litigation authority. The most common cause for error (45% of cases) identified was delayed or failed diagnosis, followed by delayed or failed treatment in 15% of cases. Communication errors were highlighted in 4% of their cases; however, poor communication plays a role in other categories, including *inappropriate medical advice resulting in delayed presentation*, which was noted to be the cause of error in 5% of cases. [22] Communication issues were highlighted as a significant category of complaints and litigation in our study. Neonatologists and paediatricians should be trained to provide accurate and understandable *safety-netting* advice to parents and caregivers, including when to seek help should their

child unexpectedly deteriorate. This is not just relevant in accident and emergency departments when discharging a child, but also on the postnatal ward, as highlighted by Hawdon et al[13] eg. counselling parents about poor feeding is vital in picking up serious medical problems such as hypoglycaemia, sepsis and jaundice. Effective safety-netting not only helps parents and caregivers feel supported and reassured, strengthening the relationship between doctor and family, but can be lifesaving if a child deteriorates acutely.

In some of the studies included in the systematic review, the exact costs incurred for financial settlement were not always reported. In 2019/20, maternity claims (including potential neonatal cases) made up 9% of the number of clinical negligence claims received by the UK National Health Service Litigation Authority (NHS LA), but represented 50% of the £4.8 billion total claim value.[23] A medical error which results in harm to a neonate is a tragedy itself and, though money received as part of a claim can be put forward to caring for the additional needs and support of the child, it is difficult to put a price on the emotional distress incurred by the family. The distress experienced by the medical professional involved is also not to be underestimated: There is little data available on this subject, however facing a medical malpractice suit can be a stressful and isolating experience for the medical professional implicated.[24]

From our review, only 4 studies[13,15,16,18] reported on factors implicated in the errors resulting in complaints, there is a need for further research and reporting of factors that result in patient harm particularly in the neonatal

Confidential: For Review Only

population. Errors leading to patient harm in neonatal medicine are very rarely due to the mistakes of a single person. As quoted by James Reason,[25] it is often due to a 'systems failure', where a series of errors occur and align together, much like the *holes* in a *Swiss cheese* model of system failure, resulting in patient harm. Adopting quality improvement and listening to staff feedback when they report underlying systems failures to create a 'culture of safety' is crucial[24]. Identifying breakdowns in the usual defence systems and using clear protocols to minimise such organisational errors will also help prevent them[24]. An awareness of human factor principles and how these influence patient safety is vital

As mentioned by Fanaroff and Goldsmith[24], 'to borrow shamelessly' from mistakes made in other units, learning from these, and implementing safety measures in our units is one way to enhance patient safety. Another area to consider is staff training with an emphasis on evidence-based guidelines. Simulation training is a relatively new learning tool that has gained popularity in the paediatric training curriculum in the UK.[26] With the aid of equipment and mannikins, scenarios are staged allowing doctors and nurses to practice working together in simulated situations. This was incorporated into the neonatal teaching programme at one UK neonatal Unit,[27] addressing neonatal-specific training issues, e.g. endotracheal tube fixation, and received unanimously positive feedback from attending staff.

Strength and Limitations

Confidential: For Review Only

From our broad-based searches of different databases, we have captured data from units across the world. However, by restricting searches to articles in English, we may have missed reports from neonatal units that have not published their data in English. Due to restrictions in language, it follows then that majority of the papers included originate from developed nations. The categories of complaints between nations showed recurring themes of 'failure or error in diagnosis or treatment'. However complaints and litigation relating to communication issues do not appear to be reported in units from less developed countries. It is likely that this is underreported as we have not captured data from non English speaking countries. Perhaps an area for further study includes collecting data from both English and non-English speaking units and exploring trends and characteristics of the complaints themselves. Furthermore, we did not include the public or patients in this systematic review, however a study investigating nature of complaints and litigation would benefit from the views and input of families who have received care from such organisations.

Given the immense progression of neonatal medicine over the last few decades, we were keen to restrict our searches to articles published from the year 2000 as we believe the lessons to be learnt from before this may not necessarily be relevant to our current practice. We excluded single case reports from this study, which inevitably has impacted the total number of articles we have reviewed. By only including papers reporting complaints *indiscriminately* from whole centres or groups of organisations, we have been able to consider our data with a lower risk of selection bias.

Confidential: For Review Only

Two of the papers[14,15] included reported on complaint data collected through the Westlaw database which is a national legal database in America. It is possible that there may be some reporting duplication as the data was collected during overlapping years 1980-2016. Muniraman et al[14] focussed on complaints relating to peripartum or immediate post-delivery management of extreme preterm infants (study population included only 22-26 weeks gestation) primarily concerning neonatal resuscitation. Nguyen et al[15] studied all communication related allegations in infants born from 22-36 weeks of gestation. This included complaints relating to resuscitation but also reported on complaints relating to other clinical categories including procedural complications, infection and retinopathy of prematurity. We made the decision to include both papers as complaints relating to several different clinical areas which would be more reflective of complaints received by all neonatal units.

Conclusions [84 words]

The most frequent categories of complaint reported in our systematic review were delay in/incorrect diagnosis and delay in/incorrect treatment. We would recommend that it is valuable for individual organisations to analyse and publish their complaints data. The aim is to pick up similar or recurring themes for complaints and this could

Confidential: For Review Only

contribute to a shared learning and appreciation of human factors that are often at play. This would help structure future training and practice in a way that champions patient safety within the discipline.

[164 words]

What is already known about this topic?

- In 2019/20, maternity claims (including neonatal cases) made up 9% of the number of clinical negligence claims received by the NHS litigation authority, but represented 50% of the total claim value. [23]
- Analysis of successful litigation claims in childhood fatalities over a 7 year period in England (in the paediatric population) showed that the top reasons included delay in diagnosis followed by delay in treatment.[22]
- In America, 60% of NICU physicians who had been in practice for over 15 years had experienced a case of malpractice litigation[7]

What does this study add?

- Communication related allegations are a significant cause of complaints. This includes communication during preterm counselling, resuscitation, during a NICU admission as well as in situations on the post-natal ward.

Confidential: For Review Only

- Factors implicated for clinician's errors that resulted in parental complaint were inadequate supervision of junior staff, work culture and hierarchy resulting in a fear of asking for help, not listening to families' concerns and system failure.

Additional Figure 1 to add to article (legend)

Figure 1: Prisma diagram[8] depicting the stages of our systematic review

Confidential: For Review Only

References

1 RCPCH. National Neonatal Audit Programme Annual Report 2020-on 2019 data. 2020. https://www.rcpch.ac.uk/sites/default/files/2020-11/nnap_report_2020_final_pdf2.pdf (accessed 10 Jan 2021).

2 Costeloe KL, Hennessy EM, Haider S, *et al.* Short term outcomes after extreme preterm birth in England: Comparison of two birth cohorts in 1995 and 2006 (the EPICure studies). *BMJ (Online)* 2012;**345**. doi:10.1136/bmj.e7976

3 Court B, Way W. Parents ' experiences of neonatal care: A report on the findings from a national survey. *Picker Institute Europe* 2011.

4 Shaw C, Connabeer K, Drew P, *et al.* Initiating end-of-life decisions with parents of infants receiving neonatal intensive care. *Patient Education and Counseling* 2020;**103**. doi:10.1016/j.pec.2020.02.013

5 Moore T, Hennessy EM, Myles J, *et al.* Neurological and developmental outcome in extremely preterm children born in England in 1995 and 2006: The EPICure studies. *BMJ (Online)* 2012;**345**. doi:10.1136/bmj.e7961

6 Aladangady N, Shaw C, Gallagher K, *et al.* Short-Term outcome of treatment limitation discussions for newborn infants, a multicentre prospective observational cohort study. *Archives of Disease in Childhood: Fetal and Neonatal Edition* 2017;**102**. doi:10.1136/archdischild-2016-310723

7 Meadow W, Bell A, Lantos J. Physicians' experience with allegations of medical malpractice in the neonatal intensive care unit. *Pediatrics* 1997;**99**. doi:10.1542/peds.99.5.e10

8 Liberati A, Altman DG, Tetzlaff J, *et al.* The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate healthcare interventions: explanation and elaboration. *BMJ (Clinical research ed)* 2009;**339**. doi:10.1136/bmj.b2700

9 Aiyengar A, Morris T, Aladangady N. PROSPERO International prospective register of systematic reviews. 2020. https://www.crd.york.ac.uk/prospero/display_record.php?ID=CRD42020167053

10 Berglund S, Norman M, Grunewald C, *et al.* Neonatal resuscitation after severe asphyxia - A critical evaluation of 177 Swedish cases. *Acta Paediatrica, International Journal of Paediatrics* 2008;**97**. doi:10.1111/j.1651-2227.2008.00803.x

11 Fallahi M, Vaghefi S, Makhmalbaf M, *et al.* Analysis of Negligence in field of Neonatology in Medical Commissions Office in Tehran, Iran. *International Journal of Medical Toxicology and Forensic Medicine* 2016;**6**. doi:10.22037/ijmtfm.v6i3(Summer).10773

12 Fanos V, Tagliabue P, Greco L, *et al.* Neonatal malpractice claims in Italy: How big is the problem and which are the causes? *Journal of Maternal-Fetal and Neonatal Medicine* 2012;**25**. doi:10.3109/14767058.2011.622004

13 Hawdon JM, Beer J, Sharp D, *et al.* Neonatal hypoglycaemia: Learning from claims. *Archives of Disease in Childhood: Fetal and Neonatal Edition* 2017;**102**. doi:10.1136/archdischild-2016-310936

14 Muniraman H, Cascione M, Ramanathan R, *et al.* Medicolegal cases involving periviable births from a major United States legal database. *Journal of Maternal-Fetal and Neonatal Medicine* 2018;**31**. doi:10.1080/14767058.2017.1335704

15 Nguyen J, Muniraman H, Cascione M, *et al.* Communication-related allegations against physicians caring for premature infants. *Journal of Perinatology* 2017;**37**. doi:10.1038/jp.2017.113

16 Rennie JM, Beer J, Upton M. Learning from claims: Hyperbilirubinaemia and kernicterus. *Archives of Disease in Childhood: Fetal and Neonatal Edition* Published Online First: 2018. doi:10.1136/archdischild-2017-314622

17 Zhou L, Li H, Li C, *et al.* Risk management and provider liabilities in infantile cerebral palsy based on malpractice litigation cases. *Journal of Forensic and Legal Medicine* 2019;**61**. doi:10.1016/j.jflm.2018.11.010

18 Ashcroft B. Labour ward incidents and potential claims - Lessons learned from research. *Clinical Risk* 2008;**14**. doi:10.1258/cr.2008.080081

19 Kaempf JW, Tomlinson MW, Tuohey J. Extremely premature birth and the choice of neonatal intensive care versus palliative comfort care: An 18-year single-center experience. *Journal of Perinatology* 2016;**36**. doi:10.1038/jp.2015.171

20 Mangurten HH, Angst DB, See C, *et al.* Professional liability in a neonatal intensive care unit: A review of 20 years' experience. *Journal of Perinatology* 2000;**20**. doi:10.1038/sj.jp.7200364

Confidential: For Review Only

21 Donn SM, Fanaroff JM. Medico-legal implications of hypothermic neuroprotection in the newborn. Journal of Neonatal-Perinatal Medicine. 2018;**11**. doi:10.3233/NPM-181792

22 Sen G, Keene J, Raine J. An analysis of successful litigation claims in childhood fatalities in England. *European Journal of Pediatrics* 2012;**171**. doi:10.1007/s00431-012-1793-6

23 National Health Service. Litigation Authority. *NHS Resolution Annual report and accounts 2019/20*.

24 Fanaroff JM, Goldsmith JP. The most common patient safety issues resulting in legal action against neonatologists. Seminars in Perinatology. 2019. doi:10.1053/j.semperi.2019.08.010

25 Reason J. Human error: Models and management. British Medical Journal. 2000;**320**. doi:10.1136/bmj.320.7237.768

26 Grant DJ, Marriage SC. Training using medical simulation. Archives of Disease in Childhood. 2012;**97**. doi:10.1136/archdischild-2011-300592

27 Sharma A. From evidence to implementation: Introducing neonatal simulation to a tertiary neonatal centre in the UK. *Open Journal of Pediatrics* 2013;**03**. doi:10.4236/ojped.2013.31002

Confidential: For Review Only

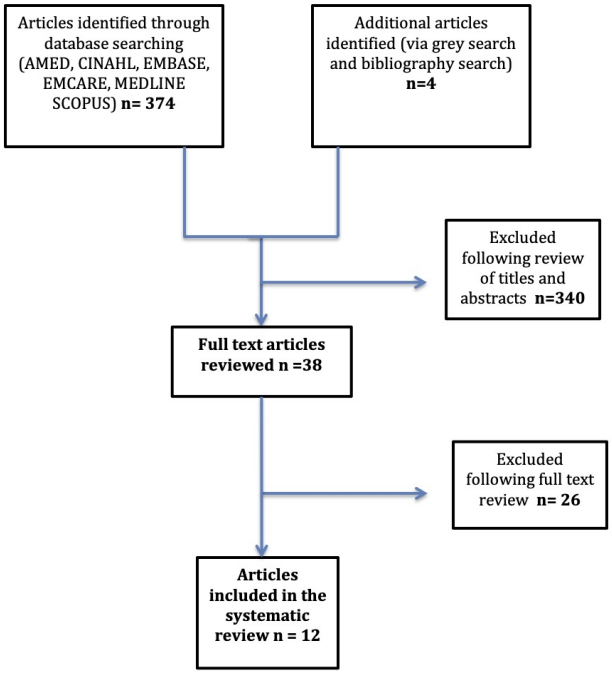


Figure 1: Prisma Diagram depicting the stages of our systematic review.

419x594mm (72 x 72 DPI)

Table 1b: Summary of the data presented in the studies that are included in the systematic review

Authors [Ref No]	Country Of Origin of the Study	Study Methodology	No of cases studied (Total)	No of complaint cases specifically involving neonatal team	Category of Clinical problem	Category of Complaint	Litigation and financial settlements
Berglund et al, 2008 [10]	• Sweden	• Retrospective review of malpractice claims (1990-2005 nationwide insurance database) • Reviewed the immediate post-delivery management in medical case notes (obstetric and neonatal) of severely asphyxiated infants following presumed malpractice in labour • Authors highlighted failure events in conjunction with neonatal resuscitation specifically based on their neonatal resus guidelines	472	177	• Neonatal resuscitation of infants born with evidence of asphyxia born after 33 weeks of gestation	• Unsatisfactory resuscitation (delay in starting artificial ventilation, intubation or chest compressions) • Unsatisfactory drug administration (lack of adrenaline or correction of metabolic acidosis) • Late initiation of resuscitation (appropriate skilled person not present at delivery or delayed in arriving) • Resuscitation not interrupted (despite >15 mins of asystole, or not spontaneously breathing for >30 mins)	• Unknown
Fallahi et al, 2016 [11]	• Tehran	• Retrospective review of malpractice cases in Tehran 2012-2014 • Data from the Medical Commissions office (National database)	53	Unknown (due to cross over between obstetric and neonatal management)	• Cerebral palsy/birth asphyxia • Respiratory distress syndrome • Sepsis • Procedure related • Seizure	• Delay in diagnosis • Delay in treatment	• Unknown

Confidential: For Review Only

Fanos et al, 2012 [12]	• Italy	• Retrospective review of malpractice claims (2005-2010) from a nationwide database in Italy • Studied neonatal claims specifically in the labour room, nursery and NICU.	661	191	• Death due to all causes including infectious, respiratory, cardiac, gastrointestinal, anoxic, other. • Permanent injury due to neurological, joint related, vision/hearing, malformative, infectious or other	• Error in diagnosis • Improper performance (incorrect action by doctor, also includes claims that are due to failure to supervise or monitor patient) • No medical misadventure (claim has legal merit, but no clear error or mishap was made by neonatologist) • Medication error	• Unknown
Hawdon et al, 2016 [13]	• United Kingdom (UK)	• Retrospective review of cases from NHS Litigation Authority Claims Management database (2002-2011) • Data extracted from the national database, letters of complaints and responses	28	28	Hypoglycaemia in infants >36/40 gestation	• Failure to appropriately monitor glucose levels despite risk factors • Early discharge without feeding support • Inadequate safety netting/discharge plan • Failure to take account of maternal concerns • Delayed testing • Missed diagnosis • Delayed referral to medical team once diagnosis made • Delayed treatment	• 25 cases damages paid • 4 cases defence only costs paid • 1 case no cost
Muniraman et al, 2017[14]	• United States of America (USA)	• Retrospective review of cases from Westlaw database (1980-2016) • National legal database • Review of medical and legal notes	15	7	Peripartum and immediate postdelivery management of infants born between 22+0-25+6 gestation	• Conflicts on resuscitation contrary to parental directives. • Miscommunication between families and clinicians.	• All cases were legally handled • 4 case verdicts in favour of professionals • 2 unknown verdicts • 1 case withdrawn

Confidential: For Review Only

Nguyen J et al, 2017 [15]	• USA	• Retrospective review of cases from Westlaw database (Jan 1975- Aug 2016) • National legal database • Review of medical and legal notes	167	54	• Procedure complication (24%) • Retinopathy of prematurity (19%) • Resuscitation (13%) • Infection (13%) • Other (31%)	• Communication related allegation (35%- 19/54) • Of above, miscommunication specifically between clinician and family (74%) including lack of informed consent, lack of disclosure, lack of anticipatory guidance and management against parental wishes	• Unknown
Rennie et al 2018 [16]	• UK	• Retrospective review of cases from NHS resolution database (2001-2011) • Data extracted from the national database, letters of claim and response	20	20	Neonatal jaundice and kernicterus	• Delay in diagnosis (measuring serum bilirubin levels despite appearing visibly jaundiced and having risk factors for neonatal jaundice e.g., family history and presence of rhesus incompatibility with positive DAT cord test n= 8) • Lag time between recognition and readmission (range between 26-102 hours) n= 15 • System failures (lack of NICU beds and inappropriate wait times in A+E resulting in delayed treatment with phototherapy) n=2	• Total costs of handling all claims = £150.5 million
Zhou et al, 2019 [17]	• China	• Retrospective analysis of cerebral palsy malpractice claims collected from 1999-2017 • National database	400 (Data on errors available in 281 cases)	199	Cerebral Palsy	• Failure of timely and effective neonatal resuscitation • Improper handling of neonates (lack of observation, inadequate treatment or poor diagnosis) • Medical management error (lost or tampered medical records, poorly documented notes, clinician without proper qualification)	• Difficult to calculate for cases specific to neonatal error. Mean value of compensation of service provider was $73, 506

Confidential: For Review Only

Ashcroft B, 2008 [18]	• UK	• Retrospective review of babies that had been admitted to the Neonatal unit with severe birth asphyxia • Collected over a set time period (Feb 2001-March 2002) from 7 maternity units (Regional data) • Cognitive interviewing technique of staff involved the data from the interviews and case notes were presented to an expert panel who applied the Bolam test to identify acceptable standard of care	37	26	Birth asphyxia requiring neonatal intensive care input	• Delay in resuscitation	• Unknown
Kaempf et al, 2016 [19]	• USA	• Retrospective review of live births in single centre (April 1996- Dec 2013) of infants born at 22+0- 26+6 to identify resuscitation and outcome • Highlight any cases with parental complaints or dissatisfaction expressed during counselling or formally via legal department	606	3	Management of extreme preterm infant against parental directive	• Lack of informed consent as parents felt they were pressured into making a decision without fully understanding the periviability counselling • Decision for palliative comfort care chosen by the parents was reversed • Inconsistent communication with healthcare team	• Unknown
Mangurten et al, 2000[20]	• USA	• Retrospective review of cases (1972-1992) • Single centre – tertiary NICU • Review of medical and legal notes	31	31	Seizure disorder (55%), hypoglycaemia (35%), pneumothorax (29%), HIE (23%), meconium aspiration (23%), asphyxia neonatorum (19%), IVH (16%), cerebral haemorrhage (16%), infantile apnoea (13%), hypotension (10%)	• Treatment error or delay (48%) • Missed/delay in diagnosis (16%) • Equipment misuse (6%) • General improper care (30%)	• 7 cases dismissed • 19 out of court settlement paid • 4 trials then damages paid • 1 unknown

Confidential: For Review Only

Donn and Fanaroff, 2017 [21]	• USA	• Review article with 2 actual legal cases presented highlighting medicolegal issues around offering neuroprotective hypothermia	2	2	Hypoxic ischaemic injury (HIE)	• Violation of the standard of care by not offering neuroprotective hypothermia (NPH) for suspected HIE • Offering NPH in an infant with an unknown cause of neurological injury leads to presupposition of an acute intrapartum event (which might not be the cause of the presentation)	• Unknown

Confidential: For Review Only